

Question number	Answer	Mark
7(a)	An explanation that includes the following points: • above 60 °C the enzymes are denatured (1) • so there will be no metabolic reactions (1) • because between 50 °C to 60 °C the enzymes are working slowly (1) • because the enzymes are denaturing (1)	(4)

Question number	Answer	Additional guidance	Mark
7(b)(i)	• $\log_{10}$ of 5×10^3 and 1.3×10^5 stated (1) • denominator calculated (1) • answer (1) Example of calculation: 3.7 and 5.1 $0.301 \times 4 = 1.204$ 1 / 1.2 / 1.16	Allow ecf Award full marks for correct numerical answer without working	(3)

Question number	Answer	Mark
7(b)(ii)	B exponential	(1)

Question number	Answer	Mark
7(c)(i)	An explanation that includes the following points: • decomposition will be slower (1) • because the enzymes will be less active (1) • as there is less {heat / kinetic} energy (1)	(3)

Question number	Answer	Mark
7(c)(ii)	An explanation that includes the following points: • cytoplasm will be frozen (1) • therefore enzymes and substrates will not be able to {move / collide} (1)	(2)

